

## Calculus II Lesson 04

1) Find  $\int \frac{1}{\sqrt{x^2 - 16}} dx$

Hint: Let  $a^2 = 16$  and  $x = a \sec \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm}} ? \hspace{1cm}$  and  $dx = \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$

b)  $\int \frac{1}{\sqrt{x^2 - 16}} dx = \int \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta \quad \left( \text{write the integral } \int \frac{1}{\sqrt{x^2 - 16}} dx \text{ in terms of } \theta \right)$

c)  $\int \frac{1}{\sqrt{x^2 - 16}} dx = \underline{\hspace{1cm}} ? \hspace{1cm}$

2) Find  $\int x^3 \sqrt{x^2 - 9} dx$

Hint: Let  $a^2 = 9$  and  $x = a \sec \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm}} ? \hspace{1cm}$  and  $dx = \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$

b)  $\int x^3 \sqrt{x^2 - 9} dx = \int \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$  (write the integral  $\int x^3 \sqrt{x^2 - 9} dx$  in terms of  $\theta$ )

c)  $\int x^3 \sqrt{x^2 - 9} dx = \underline{\hspace{1cm}} ? \hspace{1cm}$

3) Find  $\int \sqrt{4+25x^2} \, dx$

Hint:  $\sqrt{4+25x^2} = \sqrt{25\left(\frac{4}{25} + x^2\right)} = 5\sqrt{\frac{4}{25} + x^2}; \quad \int \sqrt{4+25x^2} \, dx = 5\int \sqrt{\frac{4}{25} + x^2} \, dx$

Hint: Let  $a^2 = \frac{4}{25}$  and  $x = a \tan \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm}} ? \hspace{1cm}$  and  $dx = \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$

b)  $\int \sqrt{4+25x^2} \, dx = \int \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$  (write the integral  $\int \sqrt{4+25x^2} \, dx$  in terms of  $\theta$ )

c)  $\int \sqrt{4+25x^2} \, dx = \underline{\hspace{2cm}} ? \hspace{1cm}$

4) Find  $\int \frac{1}{(9+x^2)^2} dx$

Hint: Let  $a^2 = 9$  and  $x = a \tan \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm}}?$  and  $dx = \underline{\hspace{1cm}}? \cdot d\theta$

b)  $\int \frac{1}{(9+x^2)^2} dx = \int \underline{\hspace{1cm}}? \cdot d\theta \quad \left( \text{write the integral } \int \frac{1}{(9+x^2)^2} dx \text{ in terms of } \theta \right)$

c)  $\int \frac{1}{(9+x^2)^2} dx = \underline{\hspace{2cm}}?$

5) Find  $\int \sqrt{4 - 49x^2} \, dx$

Hint:  $\sqrt{4 - 49x^2} = \sqrt{49 \left( \frac{4}{49} - x^2 \right)} = 7 \sqrt{\frac{4}{49} - x^2}; \quad \int \sqrt{4 - 49x^2} \, dx = 7 \int \sqrt{\frac{4}{49} - x^2} \, dx$

Hint: Let  $a^2 = \frac{4}{49}$  and  $x = a \sin \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm}} ? \hspace{1cm}$  and  $dx = \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$

b)  $\int \sqrt{4 - 49x^2} \, dx = \int \underline{\hspace{1cm}} ? \hspace{1cm} \cdot d\theta$  (write the integral  $\int \sqrt{4 - 49x^2} \, dx$  in terms of  $\theta$ )

c)  $\int \sqrt{4 - 49x^2} \, dx = \underline{\hspace{2cm}} ? \hspace{1cm}$

6) Find  $\int \frac{1}{\sqrt{49-x^2}} dx$

Hint: Let  $a^2 = 49$  and  $x = a \sin \theta$

a)  $\frac{dx}{d\theta} = \underline{\hspace{1cm} ? \hspace{1cm}}$  and  $dx = \underline{\hspace{1cm} ? \hspace{1cm}} \cdot d\theta$

b)  $\int \frac{1}{\sqrt{49-x^2}} dx = \int \underline{\hspace{1cm} ? \hspace{1cm}} \cdot d\theta \quad \left( \text{write the integral } \int \frac{1}{\sqrt{49-x^2}} dx \text{ in terms of } \theta \right)$

c)  $\int \frac{1}{\sqrt{49-x^2}} dx = \underline{\hspace{2cm} ? \hspace{2cm}}$



